

REACT FUNDAMENTALS

1. What is React?

React is a JavaScript library used for building user interfaces. It was created by Facebook and is commonly used to build interactive web applications.

React applications are built using reusable components.

2. Components

A component is a reusable piece of UI.

For example, a web application can have:

- Navbar component
- Sidebar component
- Login component
- Button component
- UserProfile component

A React component can be written as a function.

Example:

```
function Welcome() {  
  return <h1>Hello World</h1>;  
}
```

3. Props

Props are used to pass data from a parent component to a child component.

Example:

```
function User({ name }) {  
  return <h1>Hello {name}</h1>;  
}  
  
function App() {  
  return <User name="Rinki" />;  
}
```

Here, "Rinki" is passed from the App component to the User component through props.

4. State

State is information that belongs to a component and can change over time.

When state changes, React can re-render the component.

Example:

```
const [count, setCount] = useState(0);
```

Here:

count represents the current state.

setCount is used to update the state.

5. useState

useState is a React Hook used to add state to a functional component.

Example:

```
import { useState } from "react";
```

```
function Counter() {  
  const [count, setCount] = useState(0);  
  
  return (  
    <div>  
      <p>Count: {count}</p>  
  
      <button onClick={() => setCount(count + 1)}>  
        Increase  
      </button>  
    </div>  
  );  
}
```

When the button is clicked, setCount updates the state and React re-renders the component.

6. useEffect

useEffect is a React Hook used to perform side effects.

Examples of side effects include:

- Fetching data from an API
- Updating the document title
- Setting up a timer
- Subscribing to an event

Example:

```
useEffect(() => {  
  console.log("Component rendered");  
}, []);
```

The empty dependency array means the effect runs after the initial render.

7. Event Handling

React allows developers to respond to user actions such as:

- click
- change
- submit
- mouse events

Example:

```
<button onClick={handleClick}>  
  Click Me  
</button>
```

8. Conditional Rendering

React can display different UI depending on a condition.

Example:

```
{isLoggedIn ? <Dashboard /> : <Login />}
```

If isLoggedIn is true, Dashboard is displayed.
Otherwise, Login is displayed.

9. Lists

The map() method is commonly used to render lists.

Example:

```
const users = ["Rinki", "Amit", "Neha"];  
  
users.map(user => (  
  <p>{user}</p>  
));
```

10. Summary

React provides:

- Components for reusable UI
- Props for passing data

- State for changing data
- Hooks for additional functionality
- Event handling for user interactions
- Conditional rendering for dynamic UI